1 Composition Procedure / Recomposition

Procedure/ composition process

The procedure or composition process is to be performed as follows in general.

The relevant procedure describes how to configure the required composition functions.

Generate charging order

The composition process starts, once the melting order has been received from the ERP system or by being generated from an MES work plan.

In the next step, an employee executes the function "generate charging order" (material management → composition → composition → "start composition" tab → generate charging order). This generates the charging order that is then shown in the list of charging orders including the relevant melting order. The charging order has the order status "not free". Consequently, it cannot be used/started yet at the terminal.

Start composition

The defined composition recipe including the relevant default values as well as the permitted input materials with their respective quantities is shown by the user selecting the charging order.

Composition is now started for the charging order. Consequently, the order status changes from "not free" to "reserved".

Material assignment

Once composition has started, the user can take over the bottom sump remaining in the furnace to current composition. In this case, the current bottom sump quantity will be taken over/used. The theoretical values (planned bottom sump) assumed up to now and resulting from the sequence of orders are overwritten by current information.

Then the user can assign further materials from the list of permitted materials. The user selects the required material and clicks the function "assign material" (material management --> Composition --> Composition --> tab "perform reservation" --> assign material). By double clicking the assigned material, the user can now enter the target quantity.

Subject to the added material, the list showing the selected materials is supplemented and the theoretical analysis is updated within the sample view.

Material reservation

Once material has been assigned, the materials can be reserved for the relevant charging order within the inventory (material management --> composition --> composition --> tab "perform reservation" --> reservation/perform reservation (including material).

All or only single materials included in the list "selected materials" may be reserved for the charging order.

If the reservation is performed, the material will be reserved explicitly for this charging order and cannot be used for other charging orders.

However, it is still possible to undo or cancel reservations. Consequently, all materials are separated from the charging order and removed from the list of selected materials.

It is possible to cancel the reservation for all materials or only single materials.

In case a material has already been partially consumed, the reservation can be cancelled indeed, but the consumption will be deducted from the target quantity.

Release charging order/charging list

In case composition has been performed successfully, the charging order will be released (material management --> composition --> composition --> tab "release" --> release of charging order). The status of the charging order is "prepared". The charging order can now be used/logged on to the terminal.

The release analysis is generated by the release and saved in the system as the originally suggested charging result.

Once the charging order has been released, a "charging list" (bill of material for the charging order) can be generated by the user (material management --> composition --> composition --> tab "release" --> charging list).

Employees working in production/warehouse management are provided with the charging list to provide the relevant materials.

Procedure/ charging process

Charging is performed by an employee at the terminal pertaining to the melting furnace. To do so, the following steps are performed one after the other.

- Log charging order on
    - Log the generated charging order on
- Perform charging
    - Provision of the material in compliance with the displayed charging list from the charging order.
- The melting furnace is fed with the components and charging is completed.

- A sample is taken from the melt.
- Please also note: The result of sample taking might make re-composition necessary at MOC.
- The melt is cast. This completes the charging process.

Analysis of sampling and re-composition

The composition function enables viewing of sample results and, if necessary, to perform recomposition.

The below entries can be found in this detailed application.

Release analysis

The release analysis is the original composition result after composition has been first released and, as a result, it is the first theoretical analysis. The release analysis provides the original default values for charging and is saved/frozen as the initial status.

Theoretical analysis

The theoretical analysis first shows the current status of the used materials in relation to how the chemical make-up from the composition recipe has been achieved. Therefore, the theoretical analysis represents a default value at first.

This default value, however, is constantly recalculated, e.g. if re-composition is performed after sampling and, as a result, further materials are added to the melt.

Consequently, calculation of the current, theoretical analysis is always based on the results of actual values that are currently available in the system after sampling and not on the release analysis. This means:

- the current sample and its point in time as well as the sample weight are determined
- the actual values of the characteristics/elements of this sample are determined
- the target quantities that need to be recharged are determined (after taking the sample)
- the target material (including its make-up) is now added to the current sample's material (including its make-up) and, based on this, the composition of the theoretical analysis is calculated.

Samples (analysis based on sampling)

A sample is taken at the melting furnace. The chemical composition of the sampling is transferred to MES. The result is shown by the entry "sample" within the composition function.

If required, the user can reblend composition and add further materials. The bottom sump cannot be taken into account with recomposition. The already used bottom sump has been consumed along with the release analysis/released composition and therefore frozen.